

WayPoint

Voice-controlled indoor navigation for powered wheelchairs

TU Delft EEMCS demo (main corridor near entrance and lifts) • Authors: Ansh Kaushal; Dyorno Pavion; Ethan Creeze; Guido Nuijt; Omar Shousha; Rudrh Kapoor

Created value: more independent indoor trips for users who cannot use a joystick reliably; potential reduction of routine escort time for staff (measured in pilots).

1. Name of the product / idea

WayPoint is a plug-in voice interface that lets a powered wheelchair user speak a destination *keyword* and have the chair navigate there autonomously within a mapped indoor area. A map is shown on-screen with the available keywords; users select by reading one aloud.

2. Problem (size and urgency) and solution

Problem

Some powered-wheelchair users cannot operate a joystick consistently due to limited hand function, tremor, or fatigue. In institutions this often means short indoor trips require an escort, which reduces autonomy and consumes staff time.

Solution

WayPoint provides voice-first destination selection with confirmation, plus deterministic fallbacks:

- offline speech recognition for privacy and robustness,
- confirmation before movement,
- map click-to-navigate fallback,
- emergency stop via GUI and physical keyboard.

3. Main users / customers / beneficiaries (empathy)

Primary user: a person who can decide where they want to go but cannot use a joystick reliably.

Secondary users: OTs, nurses, assistive-technology staff who configure and supervise first use.

Buyer (first realistic): rehab centres and care facilities (pilotable because one decision covers multiple chairs).

Empathy: delays for transport can cause missed or shortened sessions; users lose control over their routine and staff time is pulled away from care tasks.

4. Use case

EEMCS demo corridor

A user wears a headset, sees a map overlay, and selects a destination by speaking a visible keyword, e.g. *Entrance, Lift A, Lift B, Tables, Pillar*. WayPoint repeats the understood destination, asks for confirmation, then sends the navigation goal.

Existing solutions and gaps

Joystick is inaccessible for some users; alternative access methods can be slower/fatiguing and require specialist fitting; staff escort creates dependence; research prototypes are not packaged for easy deployment.

What improves and why it is valuable

WayPoint enables routine indoor trips without waiting for escort availability. Value is autonomy, confidence through a consistent interaction, and better participation in scheduled activities.

Fit with the environment

The system is designed for quick reset between users, robust operation in a noisy corridor, and clear supervision controls (confirmation + e-stop + fallback).

5. Users and user groups

WayPoint targets indoor institutional environments:

- **Rehab centres (beachhead):** controlled corridors, measurable outcomes, supervised introduction.
- **Elderly care homes:** repeated routes and potential staff-time savings.
- **Hospital wards:** high value but stricter operational safety requirements.
- **Home users:** more variable environments and stronger regulatory readiness needed.
- **OEM partners:** long-term scaling path with slower decision cycles.

6. Communication plan (narrative and channels)

Narrative: “Your voice becomes the joystick for safe indoor trips.”

How users hear about it: OT networks and institutional contacts, assistive-technology communities, and short case studies.

Channels: TU Delft EEMCS demo, time-limited pilot trials with outcome reporting, direct outreach to OTs and rehab tech coordinators.

7. What is needed to bring it to a usable state?

Assumptions (pilot-scale)

Mapped indoor area with a limited labelled destination set (typically 5–15 keywords), supervised introduction, repeatable headset + laptop kit, and outcome metrics (successful trips, misrecognitions/aborts, e-stops, user/staff feedback).

Time, resources, partners

Timeline: current demo → 3–6 months pilot readiness (click-to-navigate + onboarding + logging) → 6–18 months pilots with rehab partners.

Resources: engineering time, repeatable kits, corridor access for mapping/validation, basic safety procedures.

Partners: rehab centres (OTs), platform stakeholders, institutional IT/facilities.

Two-

page synopsis describing created value, target audience, and a realistic path from demonstrator to pilot-ready system.